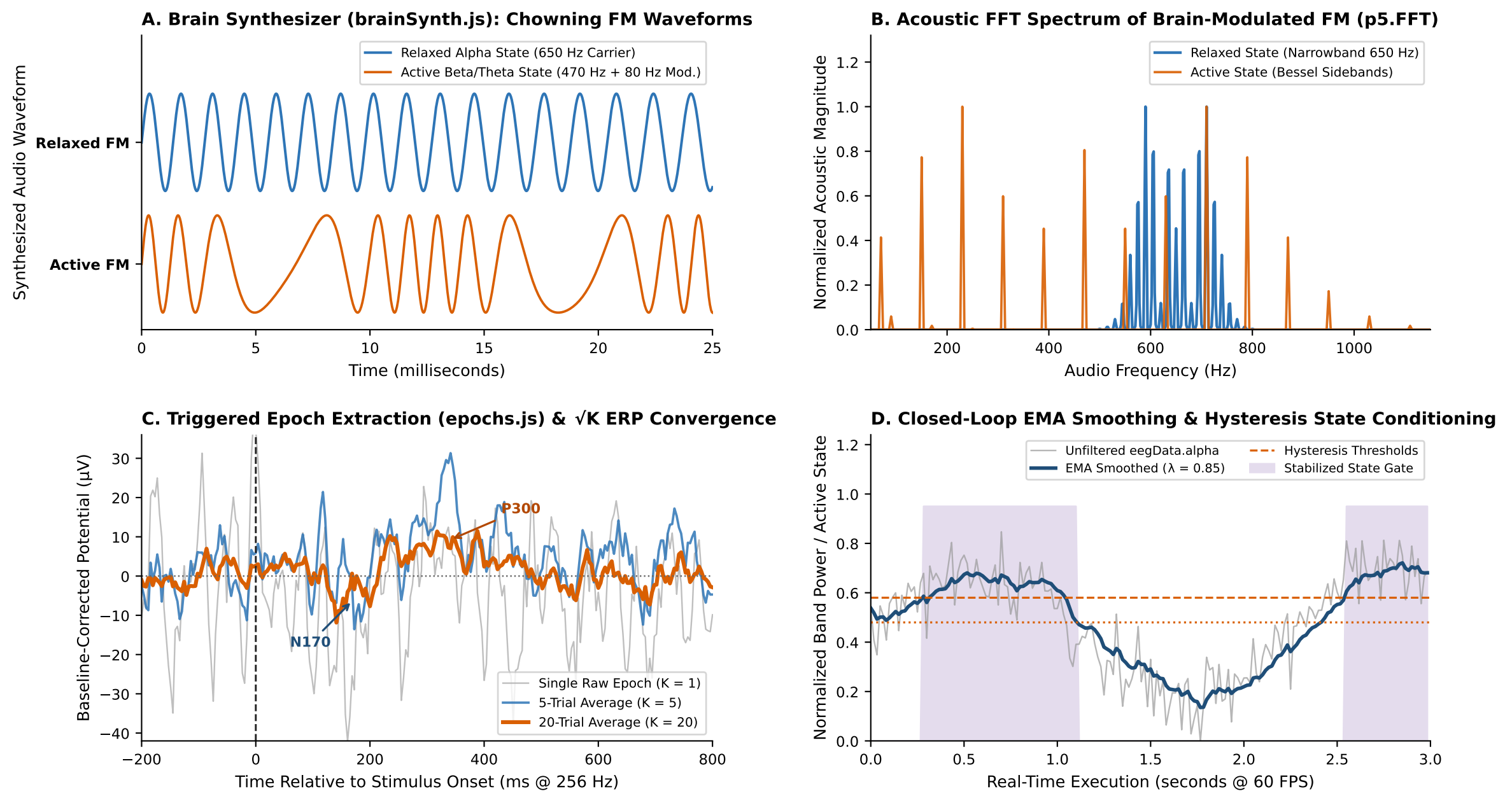

A. Brain Synthesizer (brainSynth.js): Chowning FM Waveforms
Relaxed Alpha State (650 Hz Carrier)
Active Beta/Theta State (470 Hz + 80 Hz Mod.)
Relaxed FM
Active FM
Synthesized Audio Waveform
0
5
10
15
20
25
Time (milliseconds)
B. Acoustic FFT Spectrum of Brain-Modulated FM (p5.FFT)
Relaxed State (Narrowband 650 Hz)
Active State (Bessel Sidebands)
Normalized Acoustic Magnitude
0.0
0.2
0.4
0.6
0.8
1.0
1.2
200
400
600
800
1000
Audio Frequency (Hz)
C. Triggered Epoch Extraction (epochs.js) & √K ERP Convergence
P300
N170
Single Raw Epoch (K = 1)
5-Trial Average (K = 5)
20-Trial Average (K = 20)
Baseline-Corrected Potential (μV)
30
20
10
0
−10
−20
−30
−40
−200
0
200
400
600
800
Time Relative to Stimulus Onset (ms @ 256 Hz)
D. Closed-Loop EMA Smoothing & Hysteresis State Conditioning
Unfiltered eegData.alpha
EMA Smoothed (λ = 0.85)
Hysteresis Thresholds
Stabilized State Gate
Normalized Band Power / Active State
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Real-Time Execution (seconds @ 60 FPS)